Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**6**
Electrons (Ne)	**6**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**6**
Eigenstate idx	**0 (ground state)**

Eigenstate properties

Energy E0	**-3.66870618**
Energy E1	**-2.89838147**
Gap (E1-E0)	**0.770325**
<S^2>	**-0.000000**
S	**-0.0000**
<Sz>	**0.000000**
Multiplicity	**1.0**
Spin label	**Singlet (S=0)**

Double occupancy

<D> total	**0.666395**
<D> / site	**0.111066**

RDM and GME statistics for this run (6-body clusters)

Hilbert space dim : 2^6 = 64
Number of clusters : 6
Pure clusters : 6
Mixed clusters : 0
Clusters with GME (>0) : 0

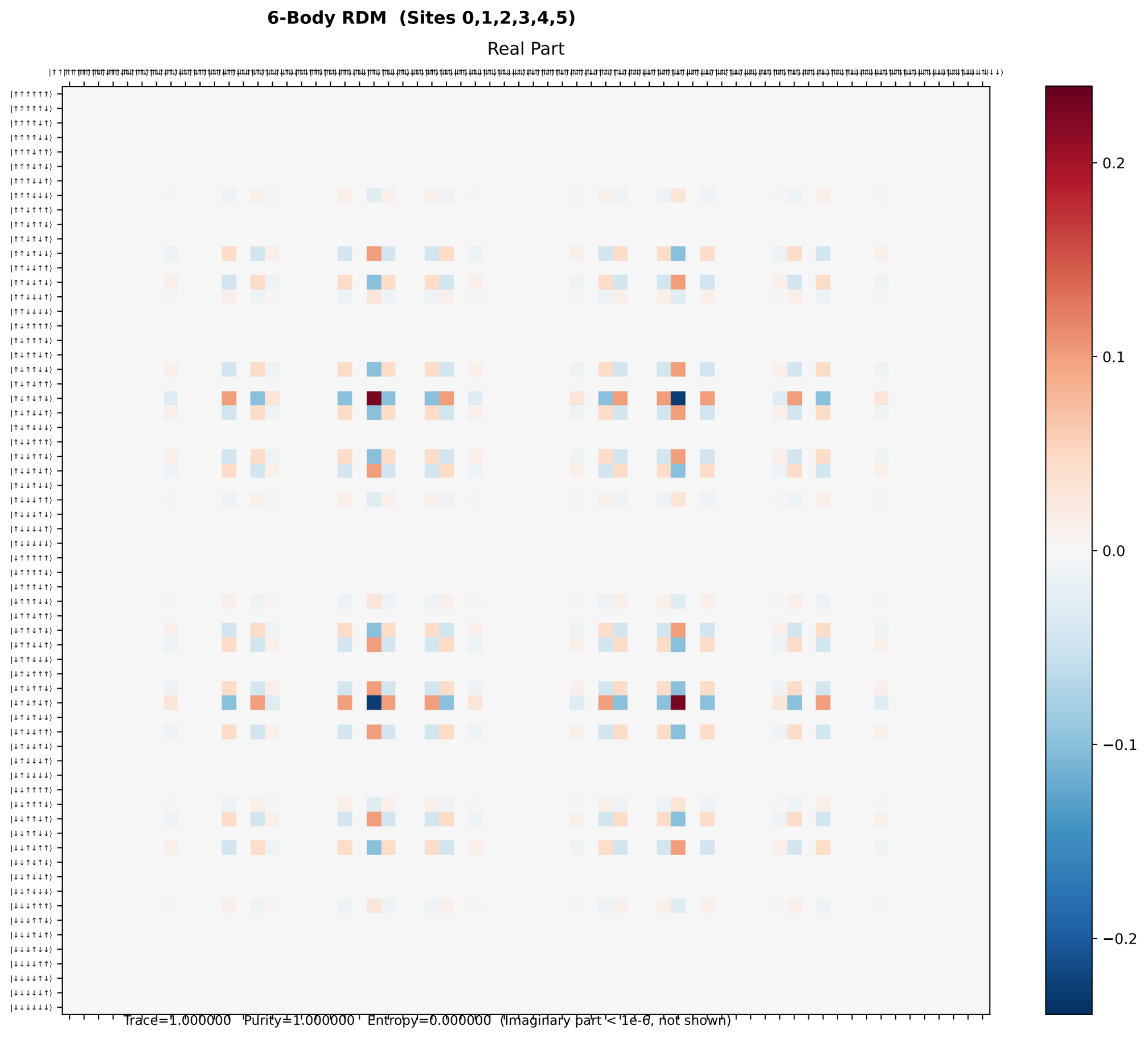
Purity gamma = Tr(rho^2) : mean = 1.000000 [min 1.000000 , max 1.000000]
Von Neumann entropy : mean = 0.000000 [min 0.000000 , max 0.000000]
Rank of rho : mean = 1.00 [min 1 , max 1]
C_GME (best estimate) : mean = — [min — , max —]

Summary: All 6-Body RDM Clusters

System: L=6, Ne=6, U=4.0, t=1.0 | E_0=-3.668706

Sites	Trace	Purity	Entropy	C_GME	C_LB	C_gcg	C_stiefel_gcg	C_stiefel	C_UB	Source	Status
(0, 1, 2, 3, 4,	1.000000	1.000000	0.000000	0.958232	–	–	–	–	–	–	GME
(1, 2, 3, 4, 5,	1.000000	1.000000	0.000000	0.958232	–	–	–	–	–	–	GME
(2, 3, 4, 5, 0,	1.000000	1.000000	0.000000	0.958232	–	–	–	–	–	–	GME
(3, 4, 5, 0, 1,	1.000000	1.000000	0.000000	0.958232	–	–	–	–	–	–	GME
(4, 5, 0, 1, 2,	1.000000	1.000000	0.000000	0.958232	–	–	–	–	–	–	GME
(5, 0, 1, 2, 3,	1.000000	1.000000	0.000000	0.958232	–	–	–	–	–	–	GME

Legend:
  Trace = Tr(rho) | Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_gcg = unitary optimization - exponential-map GCG | BEST = selected estimate
  Source: S=Stiefel, G=GCG, SG=Stiefel-GCG; PKL=reused from pkl, NEW=recomputed, MISS->NEW=pkl missing so recomputed
  C_UB = Upper bound (eigendecomp) | Status: GME=has GME | NO GME=separable | MIXED=mixed state


6-Body RDM (Sites 0,1,2,3,4,5)
Real Part
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 0,1,2,3,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-3.853794e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

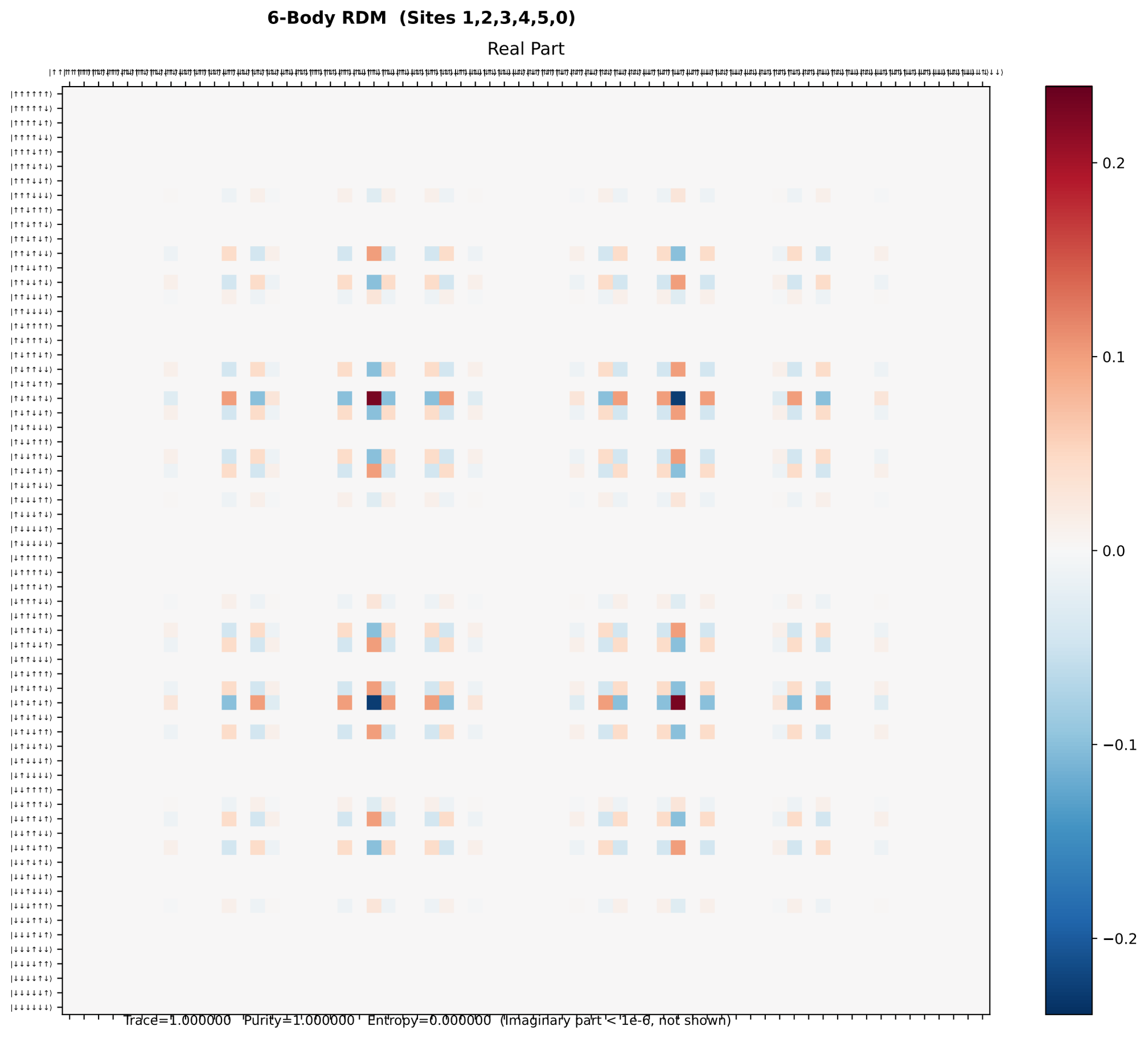
gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(0,)	(1, 2, 3, 4, 5)	0.50000000
(0, 1)	(2, 3, 4, 5)	0.54089553
(0, 2)	(1, 3, 4, 5)	0.30641624
(0, 3)	(1, 2, 4, 5)	0.31883951
(0, 4)	(1, 2, 3, 5)	0.30641624
(0, 5)	(1, 2, 3, 4)	0.54089553
(0, 1, 2)	(3, 4, 5)	0.44410365
(0, 1, 3)	(2, 4, 5)	0.33307564
(0, 1, 4)	(2, 3, 5)	0.33307564
(0, 1, 5)	(2, 3, 4)	0.44410365
(0, 2, 3)	(1, 4, 5)	0.33307564
(0, 2, 4)	(1, 3, 5)	0.20962436
(0, 2, 5)	(1, 3, 4)	0.33307564
(0, 3, 4)	(1, 2, 5)	0.33307564
(0, 3, 5)	(1, 2, 4)	0.33307564
(0, 4, 5)	(1, 2, 3)	0.44410365
(0, 1, 2, 3)	(4, 5)	0.54089553
(0, 1, 2, 4)	(3, 5)	0.30641624
(0, 1, 2, 5)	(3, 4)	0.54089553
(0, 1, 3, 4)	(2, 5)	0.31883951
(0, 1, 3, 5)	(2, 4)	0.30641624
(0, 1, 4, 5)	(2, 3)	0.54089553
(0, 2, 3, 4)	(1, 5)	0.30641624
(0, 2, 3, 5)	(1, 4)	0.31883951
(0, 2, 4, 5)	(1, 3)	0.30641624
(0, 3, 4, 5)	(1, 2)	0.54089553
(0, 1, 2, 3, 4)	(5,)	0.50000000
(0, 1, 2, 3, 5)	(4,)	0.50000000
(0, 1, 2, 4, 5)	(3,)	0.50000000
(0, 1, 3, 4, 5)	(2,)	0.50000000
(0, 2, 3, 4, 5)	(1,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.95823220
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

C_exact = 0.95823220
C_LB = 0.45569925
Gap = 0.50253294


6-Body RDM (Sites 1,2,3,4,5,0)
Real Part
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 1,2,3,4,5,0)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-2.744931e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

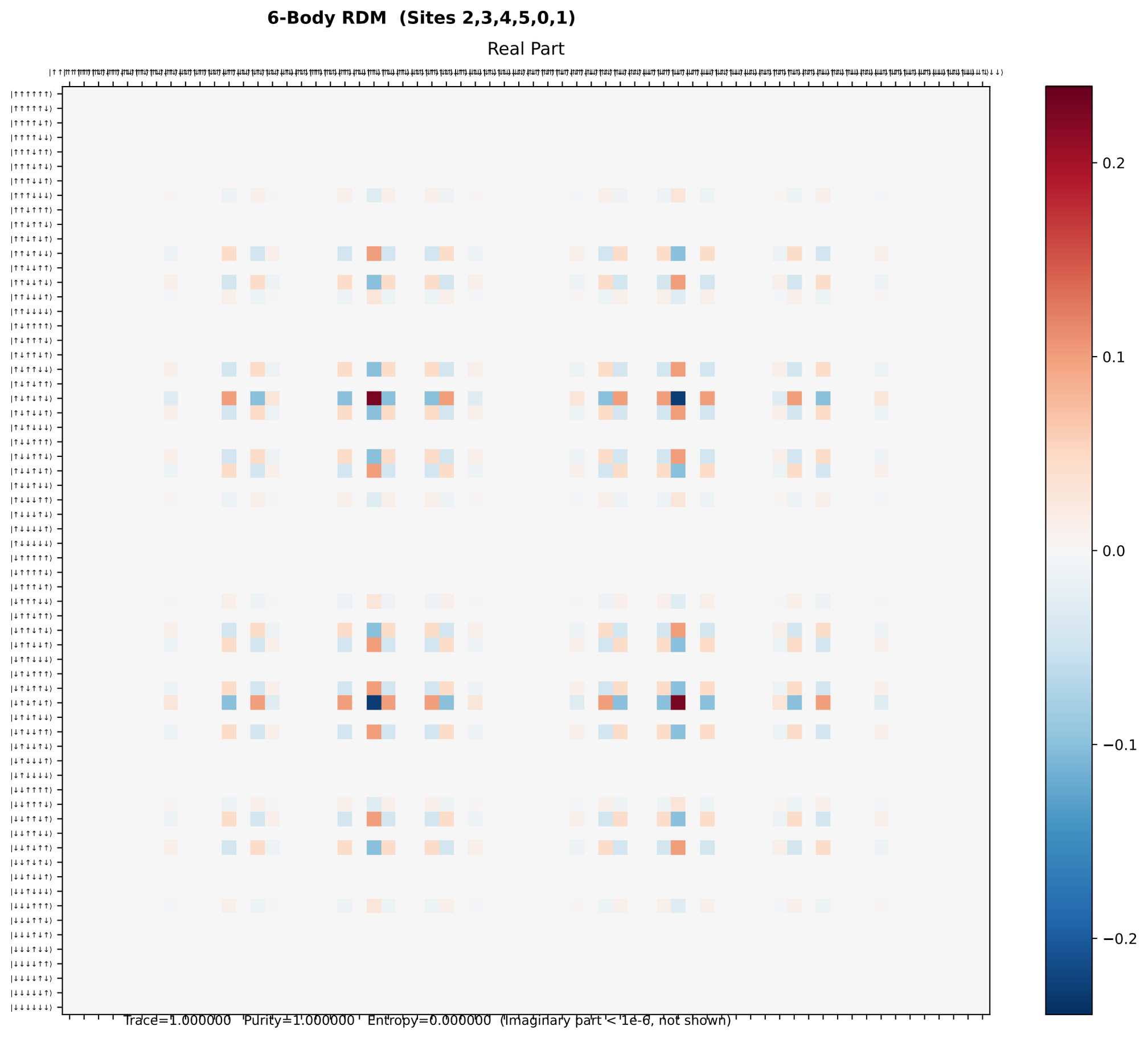
gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(1,)	(2, 3, 4, 5, 0)	0.50000000
(1, 2)	(3, 4, 5, 0)	0.54089553
(1, 3)	(2, 4, 5, 0)	0.30641624
(1, 4)	(2, 3, 5, 0)	0.31883951
(1, 5)	(2, 3, 4, 0)	0.30641624
(1, 0)	(2, 3, 4, 5)	0.54089553
(1, 2, 3)	(4, 5, 0)	0.44410365
(1, 2, 4)	(3, 5, 0)	0.33307564
(1, 2, 5)	(3, 4, 0)	0.33307564
(1, 2, 0)	(3, 4, 5)	0.44410365
(1, 3, 4)	(2, 5, 0)	0.33307564
(1, 3, 5)	(2, 4, 0)	0.20962436
(1, 3, 0)	(2, 4, 5)	0.33307564
(1, 4, 5)	(2, 3, 0)	0.33307564
(1, 4, 0)	(2, 3, 5)	0.33307564
(1, 5, 0)	(2, 3, 4)	0.44410365
(1, 2, 3, 4)	(5, 0)	0.54089553
(1, 2, 3, 5)	(4, 0)	0.30641624
(1, 2, 3, 0)	(4, 5)	0.54089553
(1, 2, 4, 5)	(3, 0)	0.31883951
(1, 2, 4, 0)	(3, 5)	0.30641624
(1, 2, 5, 0)	(3, 4)	0.54089553
(1, 3, 4, 5)	(2, 0)	0.30641624
(1, 3, 4, 0)	(2, 5)	0.31883951
(1, 3, 5, 0)	(2, 4)	0.30641624
(1, 4, 5, 0)	(2, 3)	0.54089553
(1, 2, 3, 4, 5)	(0,)	0.50000000
(1, 2, 3, 4, 0)	(5,)	0.50000000
(1, 2, 3, 5, 0)	(4,)	0.50000000
(1, 2, 4, 5, 0)	(3,)	0.50000000
(1, 3, 4, 5, 0)	(2,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.95823220
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

C_exact = 0.95823220
C_LB = 0.45569925
Gap = 0.50253294


6-Body RDM (Sites 2,3,4,5,0,1)
Real Part
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 2,3,4,5,0,1)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-4.378527e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

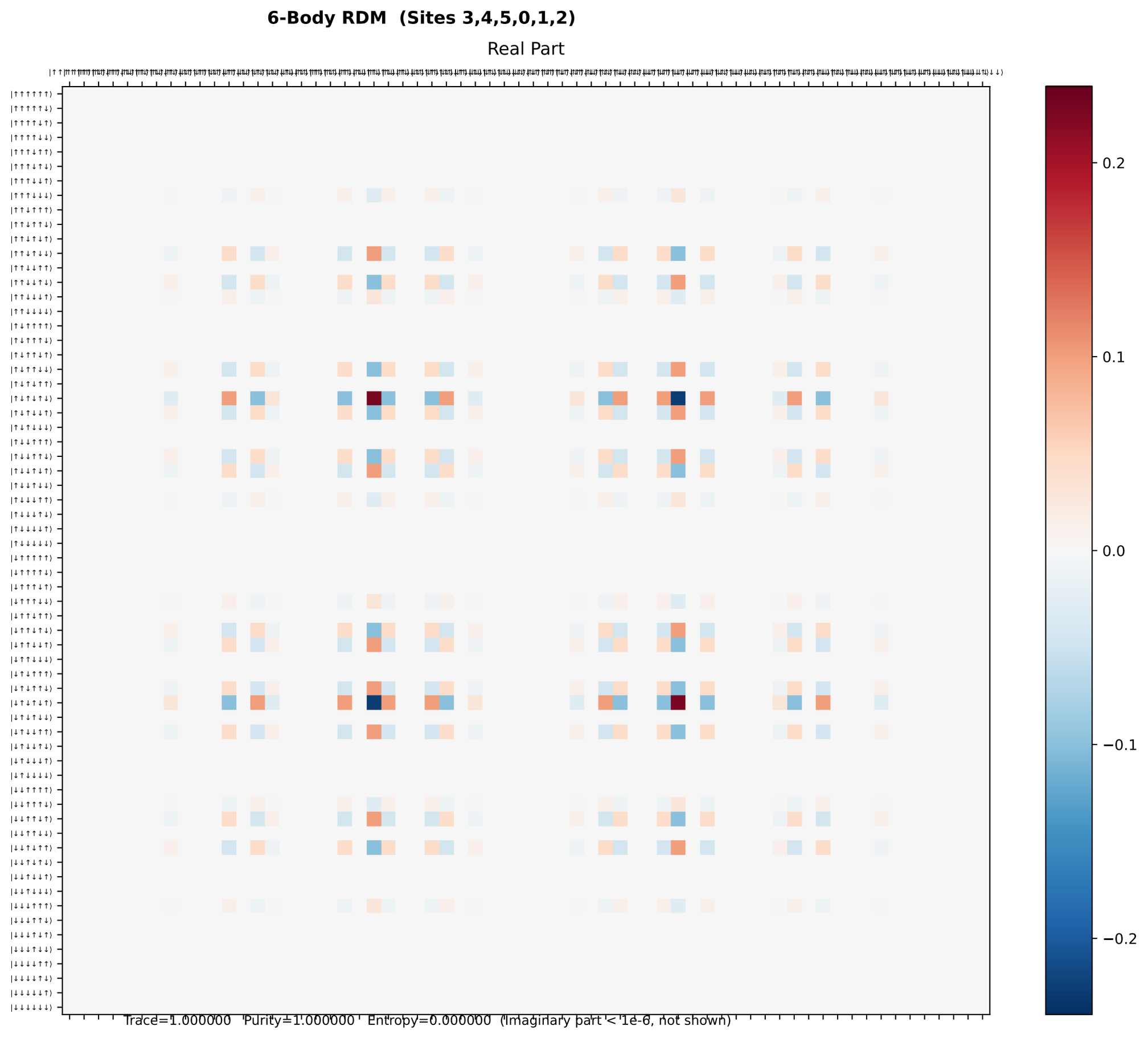
gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(2,)	(3, 4, 5, 0, 1)	0.50000000
(2, 3)	(4, 5, 0, 1)	0.54089553
(2, 4)	(3, 5, 0, 1)	0.30641624
(2, 5)	(3, 4, 0, 1)	0.31883951
(2, 0)	(3, 4, 5, 1)	0.30641624
(2, 1)	(3, 4, 5, 0)	0.54089553
(2, 3, 4)	(5, 0, 1)	0.44410365
(2, 3, 5)	(4, 0, 1)	0.33307564
(2, 3, 0)	(4, 5, 1)	0.33307564
(2, 3, 1)	(4, 5, 0)	0.44410365
(2, 4, 5)	(3, 0, 1)	0.33307564
(2, 4, 0)	(3, 5, 1)	0.20962436
(2, 4, 1)	(3, 5, 0)	0.33307564
(2, 5, 0)	(3, 4, 1)	0.33307564
(2, 5, 1)	(3, 4, 0)	0.33307564
(2, 0, 1)	(3, 4, 5)	0.44410365
(2, 3, 4, 5)	(0, 1)	0.54089553
(2, 3, 4, 0)	(5, 1)	0.30641624
(2, 3, 4, 1)	(5, 0)	0.54089553
(2, 3, 5, 0)	(4, 1)	0.31883951
(2, 3, 5, 1)	(4, 0)	0.30641624
(2, 3, 0, 1)	(4, 5)	0.54089553
(2, 4, 5, 0)	(3, 1)	0.30641624
(2, 4, 5, 1)	(3, 0)	0.31883951
(2, 4, 0, 1)	(3, 5)	0.30641624
(2, 5, 0, 1)	(3, 4)	0.54089553
(2, 3, 4, 5, 0)	(1,)	0.50000000
(2, 3, 4, 5, 1)	(0,)	0.50000000
(2, 3, 4, 0, 1)	(5,)	0.50000000
(2, 3, 5, 0, 1)	(4,)	0.50000000
(2, 4, 5, 0, 1)	(3,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.95823220
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

C_exact = 0.95823220
C_LB = 0.45569925
Gap = 0.50253294


6-Body RDM (Sites 3,4,5,0,1,2)
Real Part
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 3,4,5,0,1,2)

RDM Metrics

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-2.009816e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis

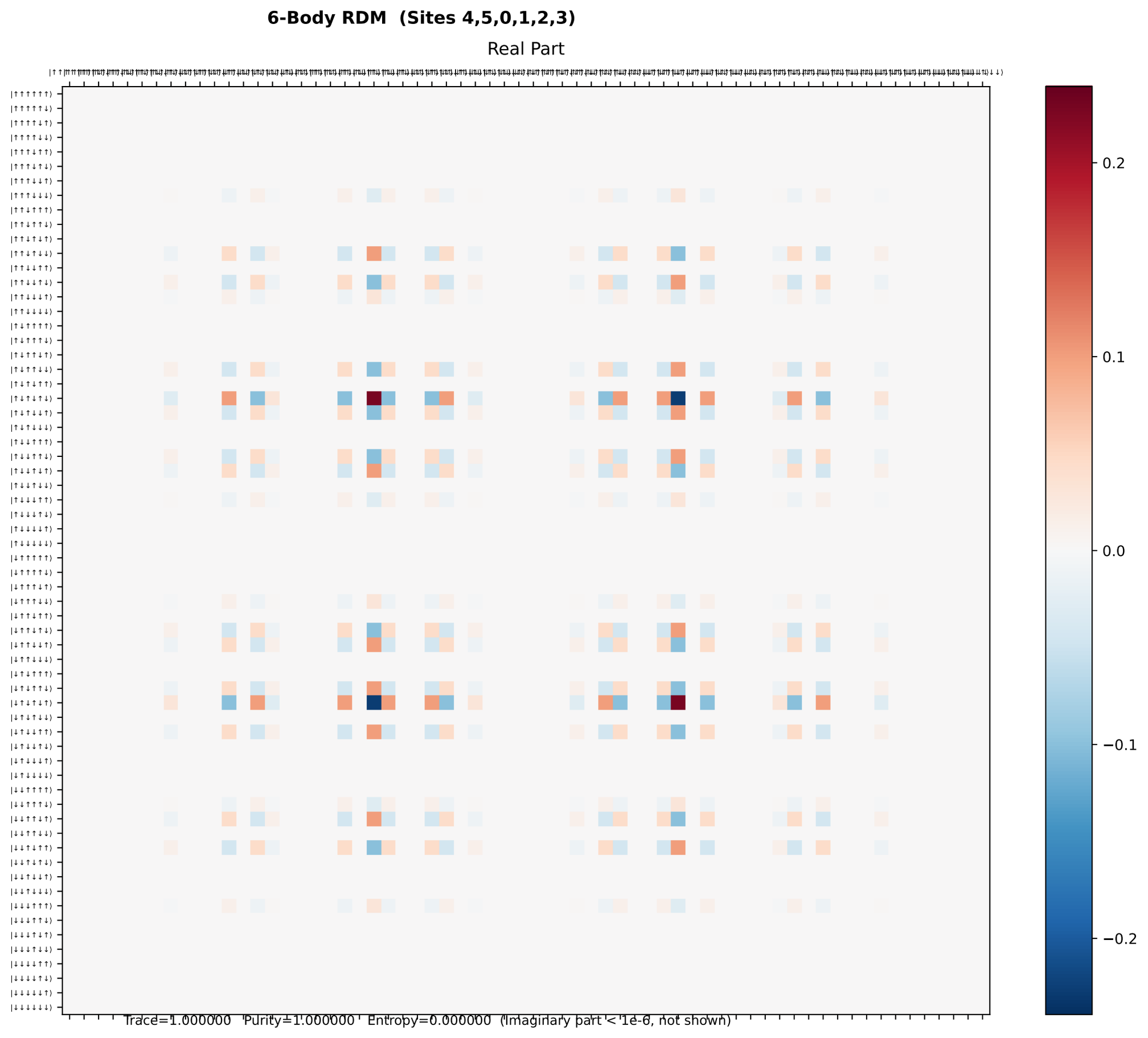
gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(3,)	(4, 5, 0, 1, 2)	0.50000000
(3, 4)	(5, 0, 1, 2)	0.54089553
(3, 5)	(4, 0, 1, 2)	0.30641624
(3, 0)	(4, 5, 1, 2)	0.31883951
(3, 1)	(4, 5, 0, 2)	0.30641624
(3, 2)	(4, 5, 0, 1)	0.54089553
(3, 4, 5)	(0, 1, 2)	0.44410365
(3, 4, 0)	(5, 1, 2)	0.33307564
(3, 4, 1)	(5, 0, 2)	0.33307564
(3, 4, 2)	(5, 0, 1)	0.44410365
(3, 5, 0)	(4, 1, 2)	0.33307564
(3, 5, 1)	(4, 0, 2)	0.20962436
(3, 5, 2)	(4, 0, 1)	0.33307564
(3, 0, 1)	(4, 5, 2)	0.33307564
(3, 0, 2)	(4, 5, 1)	0.33307564
(3, 1, 2)	(4, 5, 0)	0.44410365
(3, 4, 5, 0)	(1, 2)	0.54089553
(3, 4, 5, 1)	(0, 2)	0.30641624
(3, 4, 5, 2)	(0, 1)	0.54089553
(3, 4, 0, 1)	(5, 2)	0.31883951
(3, 4, 0, 2)	(5, 1)	0.30641624
(3, 4, 1, 2)	(5, 0)	0.54089553
(3, 5, 0, 1)	(4, 2)	0.30641624
(3, 5, 0, 2)	(4, 1)	0.31883951
(3, 5, 1, 2)	(4, 0)	0.30641624
(3, 0, 1, 2)	(4, 5)	0.54089553
(3, 4, 5, 0, 1)	(2,)	0.50000000
(3, 4, 5, 0, 2)	(1,)	0.50000000
(3, 4, 5, 1, 2)	(0,)	0.50000000
(3, 4, 0, 1, 2)	(5,)	0.50000000
(3, 5, 0, 1, 2)	(4,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.95823220
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

C_exact = 0.95823220
C_LB = 0.45569925
Gap = 0.50253294


6-Body RDM (Sites 4,5,0,1,2,3)
Real Part
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 4,5,0,1,2,3)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-3.044775e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

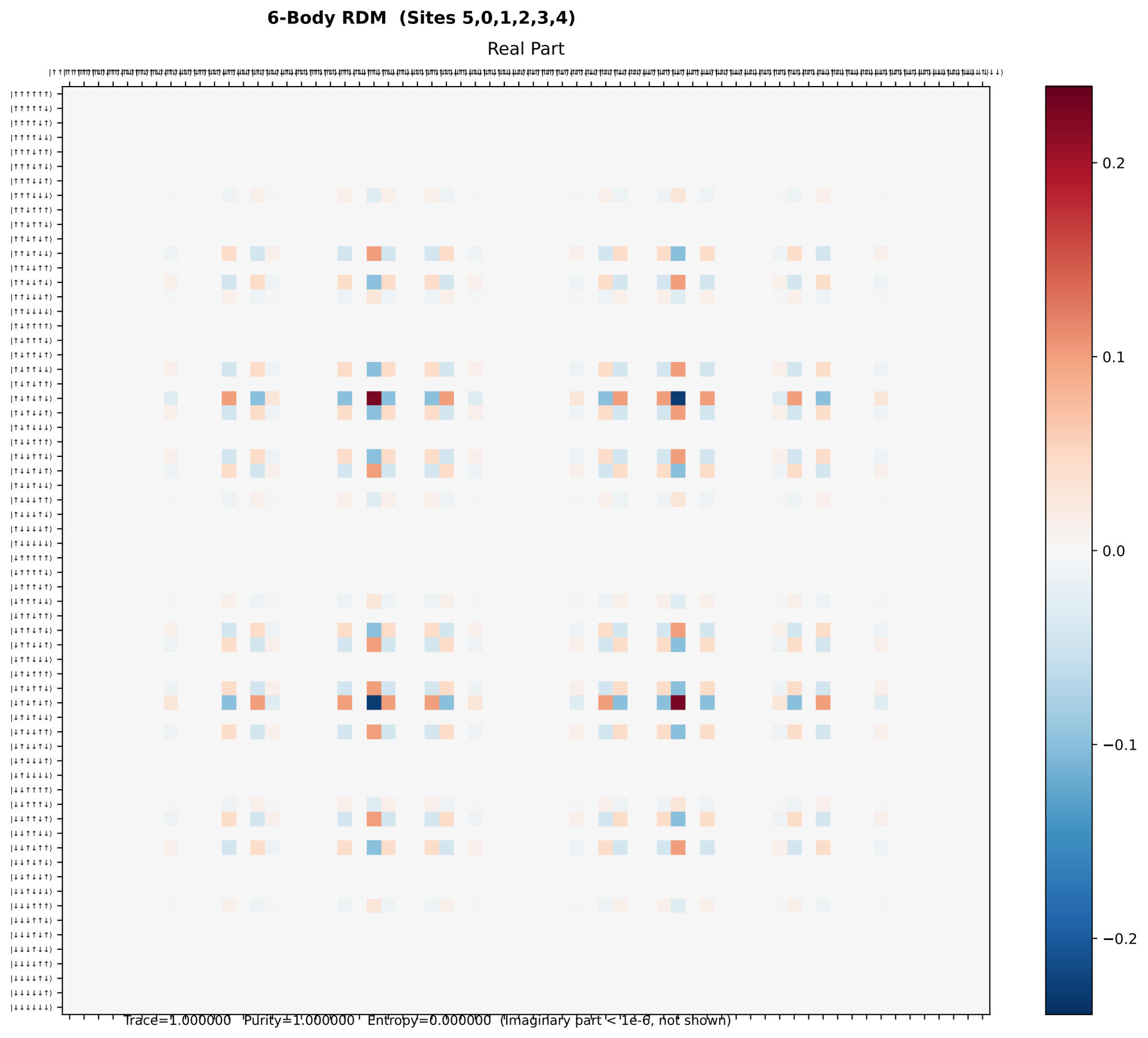
gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(4,)	(5, 0, 1, 2, 3)	0.50000000
(4, 5)	(0, 1, 2, 3)	0.54089553
(4, 0)	(5, 1, 2, 3)	0.30641624
(4, 1)	(5, 0, 2, 3)	0.31883951
(4, 2)	(5, 0, 1, 3)	0.30641624
(4, 3)	(5, 0, 1, 2)	0.54089553
(4, 5, 0)	(1, 2, 3)	0.44410365
(4, 5, 1)	(0, 2, 3)	0.33307564
(4, 5, 2)	(0, 1, 3)	0.33307564
(4, 5, 3)	(0, 1, 2)	0.44410365
(4, 0, 1)	(5, 2, 3)	0.33307564
(4, 0, 2)	(5, 1, 3)	0.20962436
(4, 0, 3)	(5, 1, 2)	0.33307564
(4, 1, 2)	(5, 0, 3)	0.33307564
(4, 1, 3)	(5, 0, 2)	0.33307564
(4, 2, 3)	(5, 0, 1)	0.44410365
(4, 5, 0, 1)	(2, 3)	0.54089553
(4, 5, 0, 2)	(1, 3)	0.30641624
(4, 5, 0, 3)	(1, 2)	0.54089553
(4, 5, 1, 2)	(0, 3)	0.31883951
(4, 5, 1, 3)	(0, 2)	0.30641624
(4, 5, 2, 3)	(0, 1)	0.54089553
(4, 0, 1, 2)	(5, 3)	0.30641624
(4, 0, 1, 3)	(5, 2)	0.31883951
(4, 0, 2, 3)	(5, 1)	0.30641624
(4, 1, 2, 3)	(5, 0)	0.54089553
(4, 5, 0, 1, 2)	(3,)	0.50000000
(4, 5, 0, 1, 3)	(2,)	0.50000000
(4, 5, 0, 2, 3)	(1,)	0.50000000
(4, 5, 1, 2, 3)	(0,)	0.50000000
(4, 0, 1, 2, 3)	(5,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.95823220
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

C_exact = 0.95823220
C_LB = 0.45569925
Gap = 0.50253294


6-Body RDM (Sites 5,0,1,2,3,4)
Real Part
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 5,0,1,2,3,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-3.538502e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(5,)	(0, 1, 2, 3, 4)	0.50000000
(5, 0)	(1, 2, 3, 4)	0.54089553
(5, 1)	(0, 2, 3, 4)	0.30641624
(5, 2)	(0, 1, 3, 4)	0.31883951
(5, 3)	(0, 1, 2, 4)	0.30641624
(5, 4)	(0, 1, 2, 3)	0.54089553
(5, 0, 1)	(2, 3, 4)	0.44410365
(5, 0, 2)	(1, 3, 4)	0.33307564
(5, 0, 3)	(1, 2, 4)	0.33307564
(5, 0, 4)	(1, 2, 3)	0.44410365
(5, 1, 2)	(0, 3, 4)	0.33307564
(5, 1, 3)	(0, 2, 4)	0.20962436
(5, 1, 4)	(0, 2, 3)	0.33307564
(5, 2, 3)	(0, 1, 4)	0.33307564
(5, 2, 4)	(0, 1, 3)	0.33307564
(5, 3, 4)	(0, 1, 2)	0.44410365
(5, 0, 1, 2)	(3, 4)	0.54089553
(5, 0, 1, 3)	(2, 4)	0.30641624
(5, 0, 1, 4)	(2, 3)	0.54089553
(5, 0, 2, 3)	(1, 4)	0.31883951
(5, 0, 2, 4)	(1, 3)	0.30641624
(5, 0, 3, 4)	(1, 2)	0.54089553
(5, 1, 2, 3)	(0, 4)	0.30641624
(5, 1, 2, 4)	(0, 3)	0.31883951
(5, 1, 3, 4)	(0, 2)	0.30641624
(5, 2, 3, 4)	(0, 1)	0.54089553
(5, 0, 1, 2, 3)	(4,)	0.50000000
(5, 0, 1, 2, 4)	(3,)	0.50000000
(5, 0, 1, 3, 4)	(2,)	0.50000000
(5, 0, 2, 3, 4)	(1,)	0.50000000
(5, 1, 2, 3, 4)	(0,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.95823220
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

C_exact = 0.95823220
C_LB = 0.45569925
Gap = 0.50253294